Switch and Router Configuration BasicsTheory

1.Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

ANS:

Create Two VLANs in Cisco Packet Tracer

Follow these steps to create VLAN 10 (Sales) and VLAN 20 (Support) on a Cisco switch.

Step 1: Add a switch

1. Open Cisco Packet Tracer.
2. Drag a 2960 switch into the workspace.
3. Click the switch and open the CLI tab.

Step 2: Enter these commands

```
enable
configure terminal

vlan 10
name Sales
exit

vlan 20
name Support
exit

end
show vlan brief
```

Step 3: Verify the VLANs

The show vlan brief command should display the following VLANs:

VLAN ID VLAN Name Status

1	default	Active
10	Sales	Active
20	Support	Active

Result: Two VLANs, Sales (10) and Support (20), have been created successfully on the switch. Take a screenshot of the CLI output for your assignment.

**2.Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Hint: Use the interface range command to select multiple ports efficiently.**

ANS:

Assign Switch Ports to VLANs in Cisco Packet Tracer

Step 1: Enter these commands in the switch CLI

```
enable
configure terminal

! Assign 3 ports to VLAN 10 (Sales)
interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit

! Assign 2 ports to VLAN 20 (Support)
interface range fa0/4-5
switchport mode access
switchport access vlan 20
exit

end
show vlan brief
```

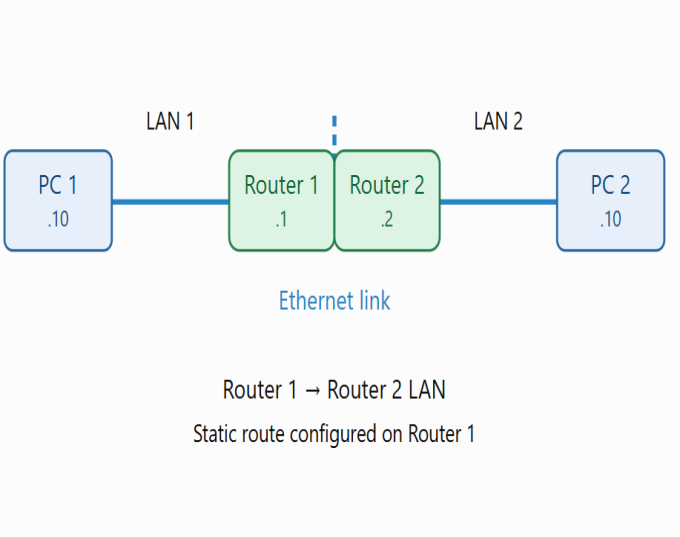
Step 2: Verify VLAN membership

Expected output:

VLAN ID	VLAN Name	Assigned Ports
1	default	Fa0/6–Fa0/24 and unused ports
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

3.Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

ANS:



Static Routing Between Two Routers in Cisco Packet Tracer

Step 1: Create the topology

Connect two routers using an Ethernet cable. Connect one PC to each router to represent the LANs.

Use these example IP addresses:

Device / Interface	IP Address	Subnet Mask
Router 1 LAN (G0/0)	192.168.1.1	255.255.255.0
Router 1 link (G0/1)	10.0.0.1	255.255.255.252
Router 2 link (G0/1)	10.0.0.2	255.255.255.252
Router 2 LAN (G0/0)	192.168.2.1	255.255.255.0

Step 2: Configure Router 1

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.1 255.255.255.252

no shutdown

exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end

show ip route

Step 3: Configure Router 2

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

exit

end

Step 4: Verify the configuration

On Router 1, enter:

show ip route

You should see a static route (S) to 192.168.2.0/24 via 10.0.0.2.

Result: Router 1 now has a static route to the LAN behind Router 2. To test end-to-end connectivity, assign IP addresses and default gateways to both PCs, then ping PC 2 from PC 1.

**4.Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.**

ANS:

Inter-VLAN Communication in Cisco Packet Tracer

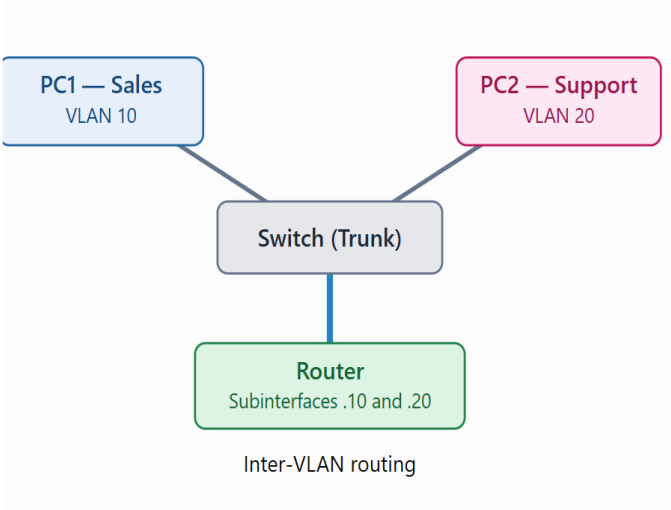
1. Scenario

- Sales VLAN 10: PC1 — IP address 192.168.10.2/24
- Support VLAN 20: PC2 — IP address 192.168.20.2/24

When PC1 tries to ping PC2, the ping fails because VLAN 10 and VLAN 20 are separate broadcast domains. A Layer 2 switch cannot route traffic between these VLANs by itself.

2. Solution: Router-on-a-Stick

Configure a router with subinterfaces for both VLANs and connect it to the switch using a trunk link.



Inter-VLAN routing

3. Example configuration

On the switch: Configure the router-facing port as a trunk (replace G0/1 with the actual port).

```
enable
configure terminal
vlan 10
 name Sales
vlan 20
 name Support
interface gigabitEthernet 0/1
 switchport mode trunk
end
```

On the router: Configure one subinterface for each VLAN.

```
enable
configure terminal

interface gigabitEthernet 0/0
 no shutdown
exit

interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit

end
```

Set the PCs' default gateways to 192.168.10.1 for Sales and 192.168.20.1 for Support.

4. Test and result

From PC1, run:

ping 192.168.20.2

Before inter-VLAN routing is configured, the ping fails. After configuring the trunk, router subinterfaces, IP addresses, and gateways correctly, the ping should succeed.

Alternative: A Layer 3 switch can also enable inter-VLAN communication using VLAN interfaces (SVIs) and IP routing.